

CENTRAL HEATING WATER REROUTE MANUAL

Surface-Mounted Two-Pipe Copper Circuit — 10-Stage Technical Execution Guide

Stage 1: Pre-Work Approvals & Governance

Prior to physical works, formal consent must be obtained from freeholders and insurers. The Gas Safe engineer validates the planned route, pipe diameter calculations, and property leasehold compliance before any drilling or material purchases occur.



Figure 1: Stage 1: Pre-Work Approvals & Governance Vector Infographic

Stage 2: Route Measurement & Inventory Mapping

Physical measurement along room perimeters. Calculates flow and return pipe runs by doubling the one-way distance and adding a 10% waste buffer. Outer diameters are verified with precision vernier calipers before purchasing.



Figure 2: Stage 2: Route Measurement & Inventory Mapping Vector Infographic

Stage 3: System Drainage & Electrical Isolation

Isolate power via fused spur electrical switch. Connect garden hose to low-level drain valve, drain liquid cleanly into buckets, and bleed radiators until static pressure reads 0.0 bar.

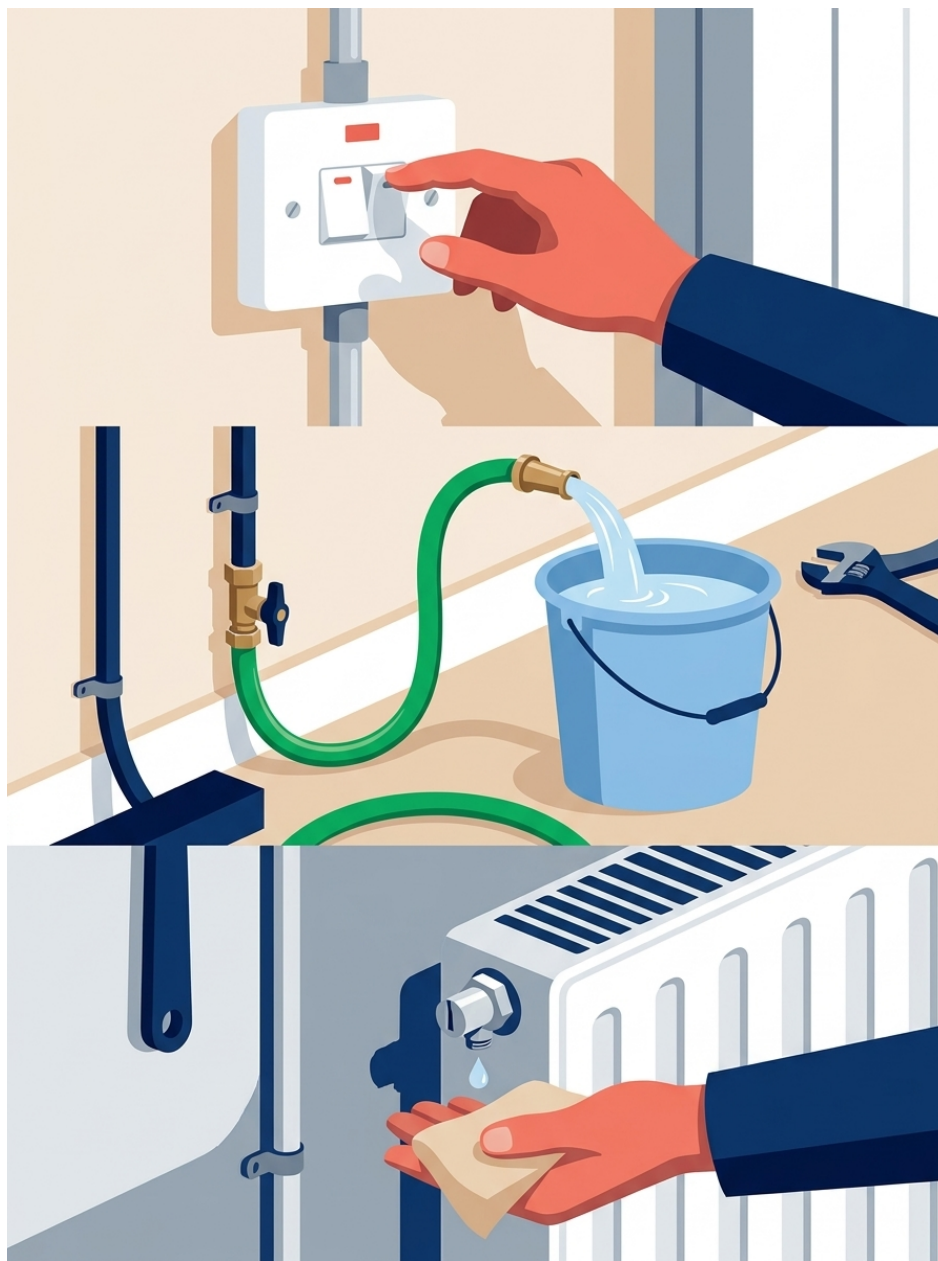


Figure 3: Stage 3: System Drainage & Electrical Isolation Vector Infographic

Stage 4: Wall Penetrations & Pipe Supports

Electronic wall scanning identifies hidden wires and studs. Clean horizontal wall penetrations are drilled above skirting boards. Rubber-lined clips secure flow and return lines while preventing acoustic vibration transfer.



Figure 4: Stage 4: Wall Penetrations & Pipe Supports Vector Infographic

Stage 5: Push-Fit Assembly — The Golden Sequence

Five mandatory assembly steps: (1) Square rotary cutting, (2) Internal/external deburring, (3) Surface cleaning, (4) Insertion depth marking, and (5) Firm push-fit seating to guarantee leak-free O-ring sealing.

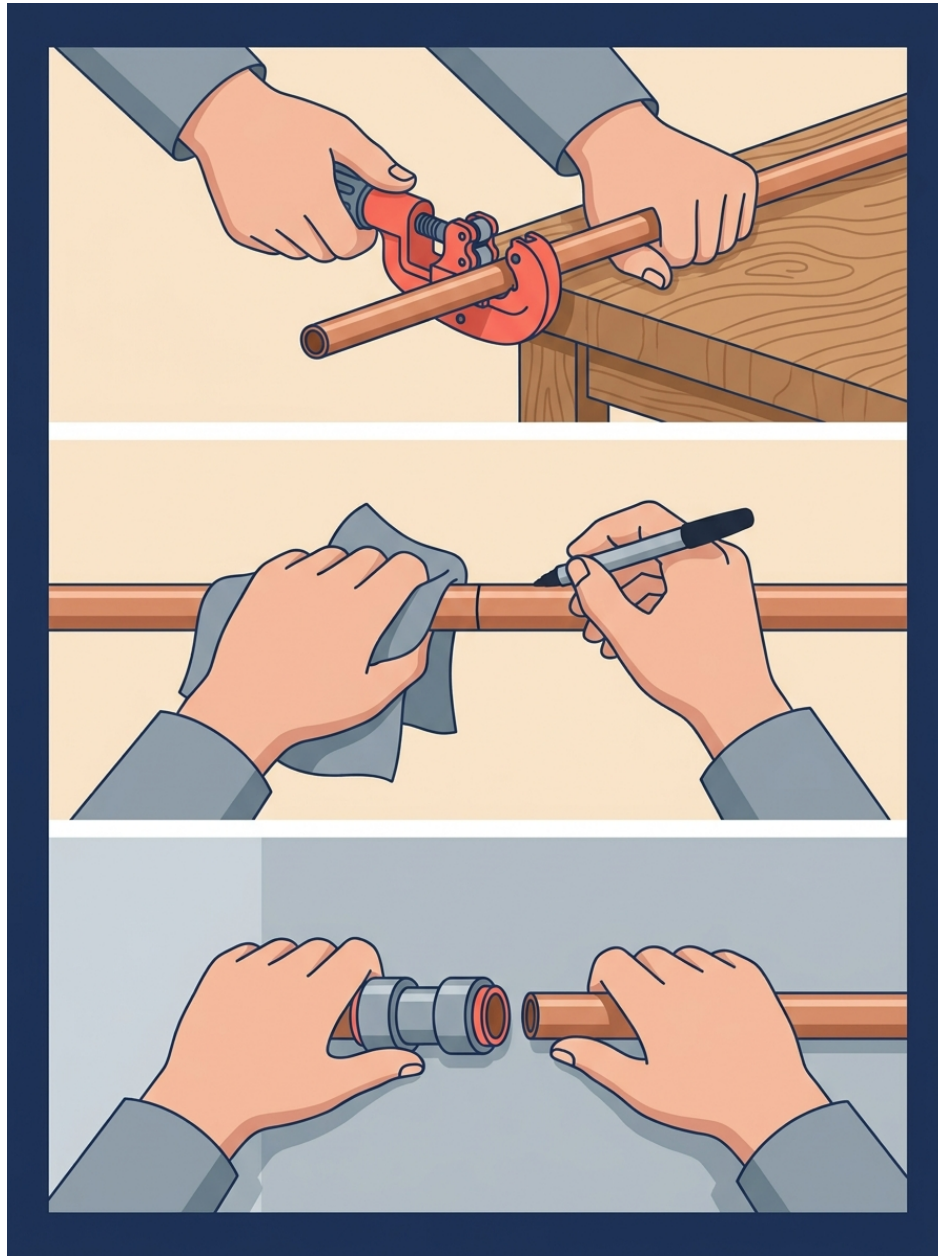


Figure 5: Stage 5: Push-Fit Assembly — The Golden Sequence Vector Infographic

Stage 6: Radiator Branchwork & Valve Splitting

Equal brass T-fittings spliced into main trunk lines to supply individual panel radiators in parallel. Upward branch lines connect directly into TRV and lockshield bottom entry valves.

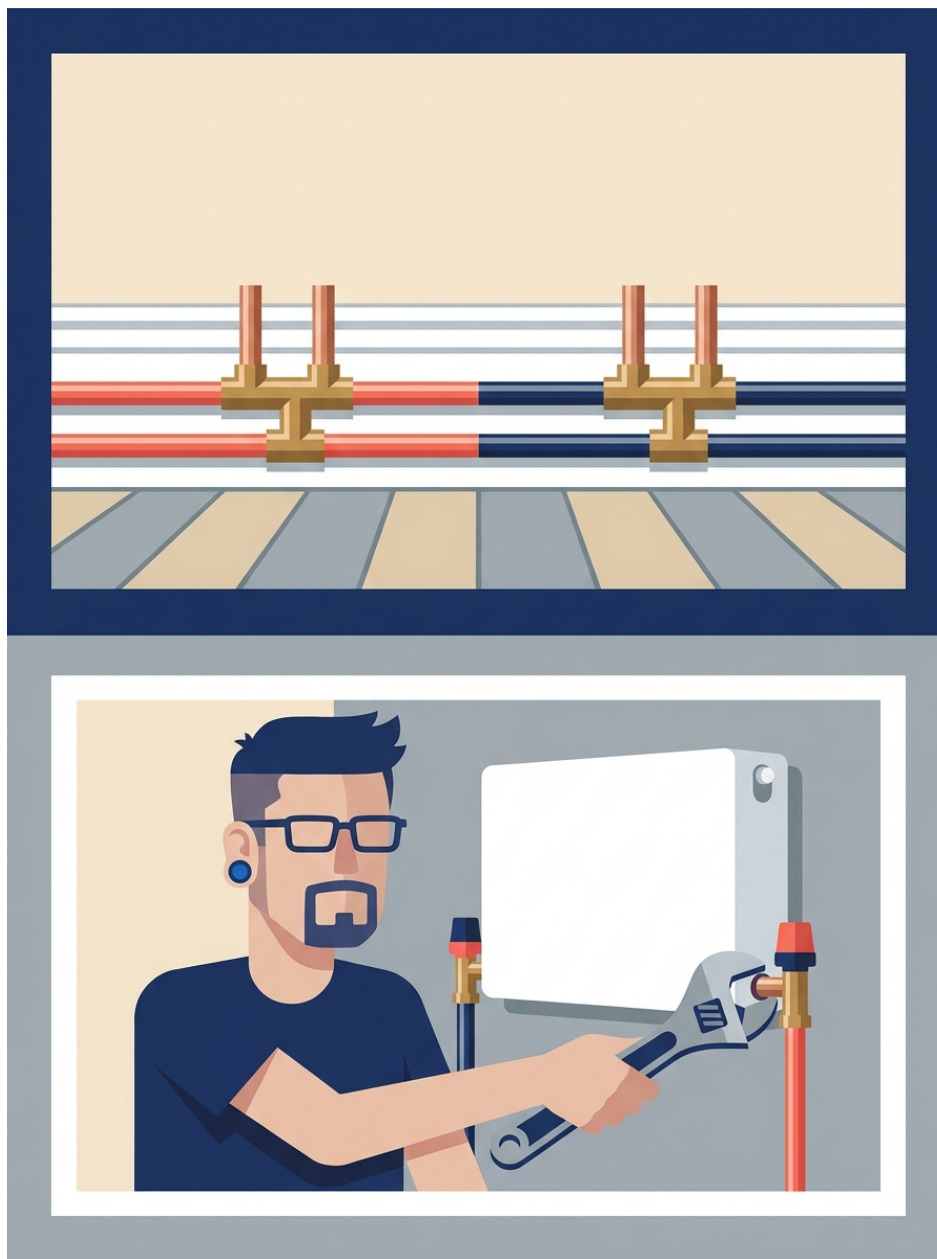


Figure 6: Stage 6: Radiator Branchwork & Valve Splitting Vector Infographic

Stage 7: Final Loop Closure & Circuit Verification

Terminal radiator connections using 90-degree elbows. Return trunk lines follow a continuous path back to the boiler location, verified by manual line tracing.

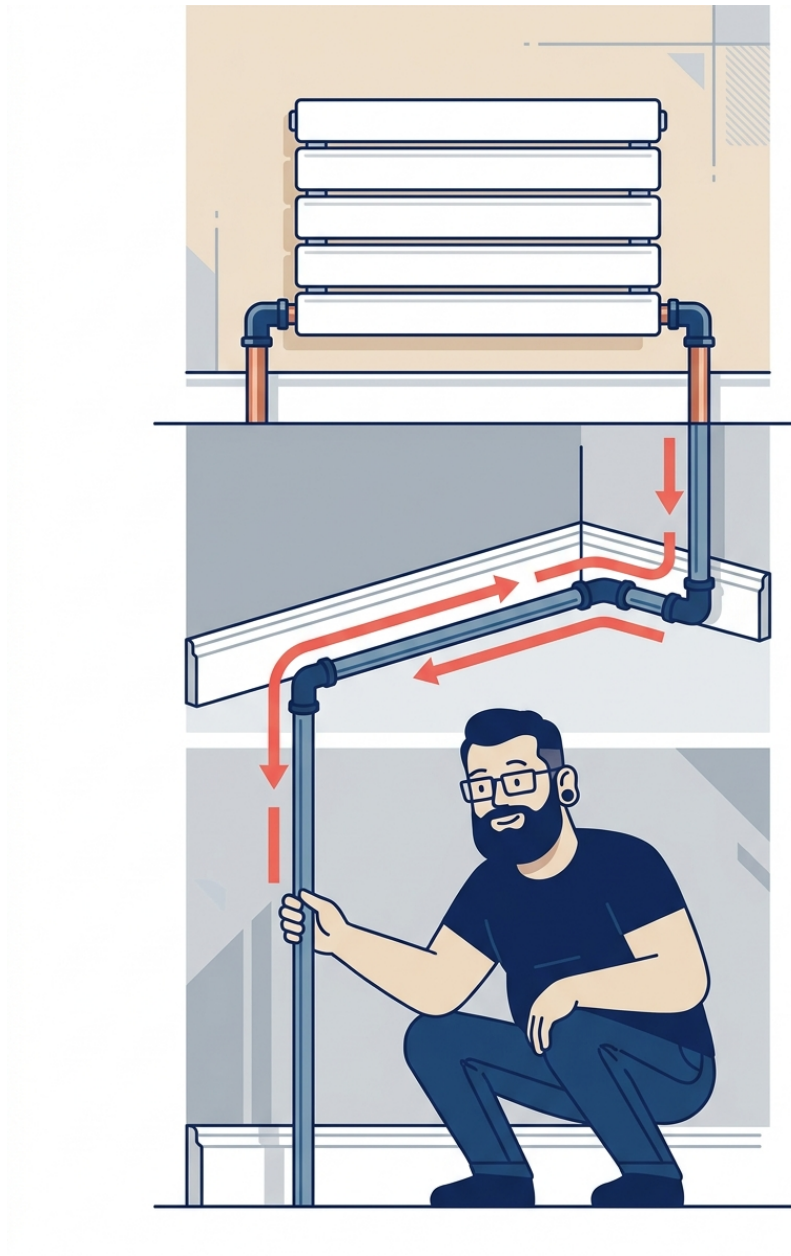


Figure 7: Stage 7: Final Loop Closure & Circuit Verification Vector Infographic

Stage 8: Gas Safe Boiler Reconnection & Handover

Strict professional boundary: A qualified, clean-shaven Gas Safe registered engineer executes underslung copper pipe reconnections to the combination boiler and signs off safety documentation.



Figure 8: Stage 8: Gas Safe Boiler Reconnection & Handover Vector Infographic

Stage 9: System Refill, Bleeding & Hydrostatic Testing

System refilled and pressurized to 1.0–1.5 bar. Air bled from radiators. Blue tissue paper leak checks applied across all push-fit joints under full thermal operating temperature.



Figure 9: Stage 9: System Refill, Bleeding & Hydrostatic Testing Vector Infographic

Stage 10: Aesthetic Boxing-In & Permanent Maintenance Access

Timber battens and removable grey MDF boxing panels installed over skirting pipework. Hinged access hatches positioned over isolation valves and major joints to ensure future access.

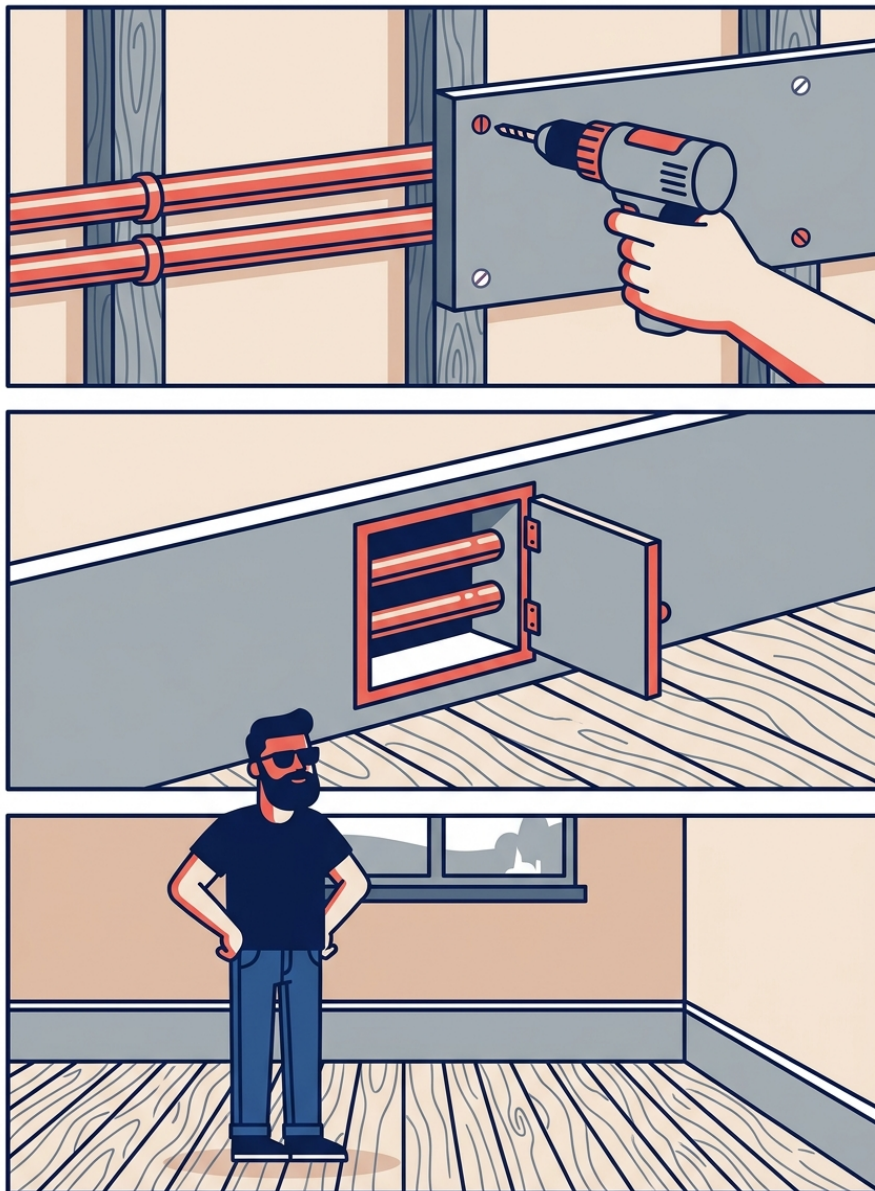


Figure 10: Stage 10: Aesthetic Boxing-In & Permanent Maintenance Access Vector Infographic